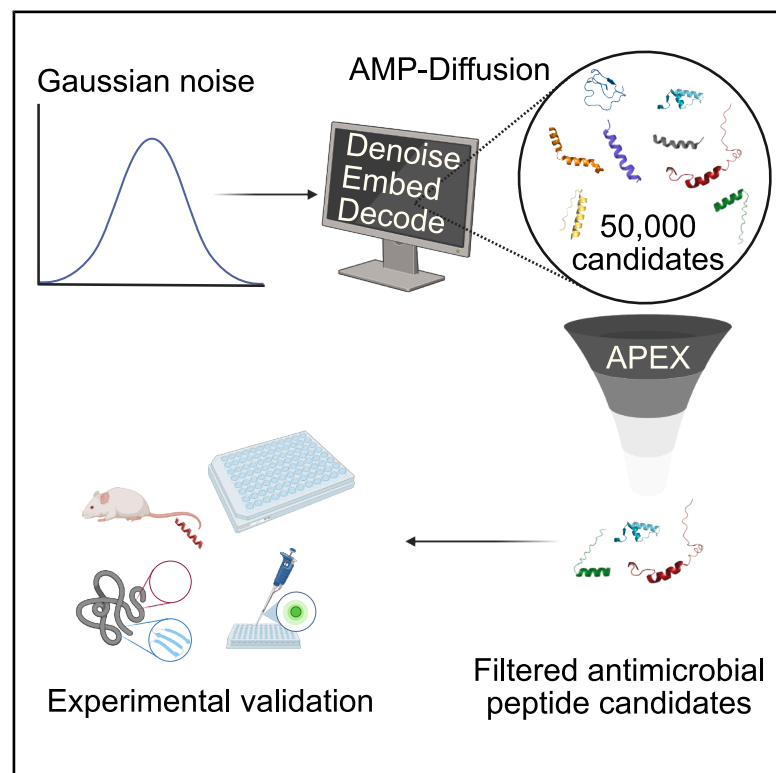


Generative latent diffusion language modeling yields anti-infective synthetic peptides

Graphical abstract



Authors

Marcelo D.T. Torres, Leo Tianlai Chen, Fangping Wan, Pranam Chatterjee, Cesar de la Fuente-Nunez

Correspondence

pranam@seas.upenn.edu (P.C.),
cfuente@upenn.edu (C.d.I.F.-N.)

In brief

AMP-Diffusion is a generative artificial intelligence (AI) framework that creates antimicrobial peptides (AMPs) using a latent diffusion model trained on protein language embeddings. Torres et al. show that AI-designed peptides exhibit potent, broad-spectrum antibacterial activity, low toxicity, and *in vivo* efficacy comparable to standard antibiotics. This work presents a scalable, experimentally validated platform for rational antibiotic discovery, with broader implications for therapeutic peptide design.

Highlights

- AMP-Diffusion designs potent peptides using latent diffusion modeling
- 76% of tested peptides kill bacteria with low toxicity
- Lead peptides reduce infections *in vivo* with no adverse effects
- Diffusion AI generates diverse, bioactive peptides *de novo*



Article

Generative latent diffusion language modeling yields anti-infective synthetic peptides

Marcelo D.T. Torres,^{1,2,3,4,9} Leo Tianlai Chen,^{5,9} Fangping Wan,^{1,2,3,4} Pranam Chatterjee,^{5,6,7,8,*} and Cesar de la Fuente-Nunez^{1,2,3,4,10,*}

¹Machine Biology Group, Departments of Psychiatry and Microbiology, Institute for Biomedical Informatics, Institute for Translational Medicine and Therapeutics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA 19104, USA

²Departments of Bioengineering and Chemical and Biomolecular Engineering, School of Engineering and Applied Science, University of Pennsylvania, Philadelphia, PA 19104, USA

³Department of Chemistry, School of Arts and Sciences, University of Pennsylvania, Philadelphia, PA 19104, USA

⁴Penn Institute for Computational Science, University of Pennsylvania, Philadelphia, PA 19104, USA

⁵Department of Biomedical Engineering, Duke University, Durham, NC 27708, USA

⁶Department of Computer Science, Duke University, Durham, NC 27708, USA

⁷Department of Biostatistics and Bioinformatics, Duke University, Durham, NC 27708, USA

⁸Present address: Department of Bioengineering and Computer and Information Science, School of Engineering and Applied Science, University of Pennsylvania, Philadelphia, PA 19104, USA

⁹These authors contributed equally

¹⁰Lead contact

*Correspondence: pranam@seas.upenn.edu (P.C.), cfuente@upenn.edu (C.d.l.F.-N.)

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THE BIGGER PICTURE Antibiotic resistance is accelerating faster than our ability to discover new drugs. Antimicrobial peptides (AMPs) are promising alternatives, but navigating the immense sequence space while meeting potency and safety constraints has been a bottleneck. Here, we introduce AMP-Diffusion, a generative artificial intelligence (AI) platform that uses latent diffusion modeling and protein language embeddings to create biologically relevant results without predefined motifs or structural priors. From 50,000 *in silico* designs, we synthesized and tested 46 peptides: 76% inhibited bacteria, including multidrug-resistant strains, while showing minimal toxicity. Two lead candidates showed *in vivo* efficacy in mice that was comparable to clinically used antibiotics. This study illustrates how generative AI can rapidly identify and optimize therapeutic peptides, offering a scalable and generalizable approach to antibiotic development. AMP-Diffusion sets the stage for future platforms that tailor peptides for specific pathogens or therapeutic targets.

SUMMARY

Generative artificial intelligence (AI) offers a powerful avenue for peptide design, yet this process remains challenging due to vast sequence space, complex structure-activity relationships, and the need to balance antimicrobial potency with low toxicity. Here, we introduce AMP-Diffusion, a latent diffusion model fine-tuned on antimicrobial peptide (AMP) sequences using embeddings from protein language models (pLMs). AMP-Diffusion enables the rapid discovery of antibiotic candidates by systematically exploring sequence space. We generated 50,000 candidate sequences, filtered and ranked them using our APEX deep learning (DL) model, and synthesized 46 top candidates. These peptides showed broad-spectrum antibacterial activity, including against multidrug-resistant strains, while exhibiting low cytotoxicity. Mechanistic studies revealed membrane permeabilization and depolarization as primary modes of action. In a preclinical mouse model, lead peptides reduced bacterial loads with efficacy comparable to polymyxin B and levofloxacin, with no detectable adverse effects. AMP-Diffusion thus presents a robust platform for designing antibiotics.

INTRODUCTION

The field of generative artificial intelligence (AI) has experienced significant advancements, with denoising diffusion

models emerging as effective tools for diverse tasks in computer vision, natural language processing, and more recently, protein design.¹ By progressively denoising data from a Gaussian noise distribution back to the original signal,



diffusion models enable the exploration of innovative strategies for designing proteins with specific structural and functional characteristics.

In protein design, diffusion-based approaches such as RFDiffusion have successfully engineered protein monomers, binders, and symmetric oligomers, leveraging their ability to explore vast conformational spaces.² Similarly, graph-based diffusion models have enabled progress in specialized areas such as antibody design and protein-ligand docking.^{3,4} By leveraging graph representations of protein structures and interactions, these models refine and optimize molecular configurations, leading to antibodies with improved binding affinities or ligands displaying enhanced docking properties.

Alongside diffusion-based methods, protein language models (pLMs) have emerged as powerful tools for protein sequence analysis and design. Models such as ESM-2,⁵ ProtT5,⁶ and others are pre-trained on large-scale datasets comprising millions of natural protein sequences. Through transformer architectures, they capture fundamental physicochemical and functional properties directly from the sequences. This capability extends to *de novo* protein design, where pLMs can generate sequences that retain functionality and bioactivity. For example, ProtGPT2⁷ and ProGen⁸ are pre-trained in an autoregressive manner, while EvoDiff employs diffusion to support both *de novo* sequence generation and scaffolding.⁹

Antimicrobial peptides (AMPs) offer a promising alternative to conventional antibiotics due to their multifaceted mechanisms of action, including membrane disruption, metabolic inhibition, and immune modulation.^{10–12} This versatility makes them potent candidates for countering antimicrobial resistance (AMR).^{13–15} Despite their potential, AMP discovery remains challenging. Machine learning (ML) and deep learning (DL) have become indispensable tools in AMP discovery and generation, accelerating research by identifying functional patterns in large datasets.^{16–26}

Given the rising threat of AMR, we hypothesized that integrating representation and generative language modeling would facilitate robust AMP generation, enabling precise control over peptide properties. In this work, we show that AMP-Diffusion, a pre-trained latent diffusion model, generates functionally potent AMP sequences by applying Gaussian noise over the pre-trained ESM-2-650M embedding space.^{6,27} Compared with other works that use latent diffusion models to design AMPs,^{28,29} AMP-Diffusion requires no protein latent-space training by directly operating the forward/reverse diffusion processes on the latent space derived from the robust and widely validated pLM ESM-2, which is considered a gold standard for sequence-based protein understanding. In addition, our denoiser architecture ensures a deeper integration with the base pLM. By reusing ESM-2's attention-layer weights in our denoiser, AMP-Diffusion is better constrained to generate sequences that are not only statistically likely but also adhere to the complex, underlying biological rules learned by ESM-2. We provide extensive ground-truth experimental validation of AMP-Diffusion-generated peptides by evaluating their structure, mechanism of action, cytotoxicity, and efficacy in animal infection models.

RESULTS AND DISCUSSION

Computational framework

We developed AMP-Diffusion, a latent-space diffusion model designed for AMP generation.²⁷ The model generates functional AMPs by applying a diffusion process in the latent space of ESM-2 embeddings, progressively adding and then removing noise to generate peptide sequences. AMP-Diffusion was trained on a compiled dataset of AMPs from DRAMP 3.0,³⁰ APD3,³¹ and DBAASP³² databases (Dataset S1).

Our evaluation shows that AMP-Diffusion-generated peptides closely match experimentally validated AMPs in terms of pseudo-perplexity and amino acid residue diversity. These generated sequences also display physicochemical properties comparable to those of naturally occurring AMPs, suggesting their biological plausibility and potential for empirical validation.²⁷ Model derivation and details can be found in Chen et al.²⁷ For this study, we hypothesized that our pre-trained AMP-Diffusion model would enable the generation of functionally potent AMPs.

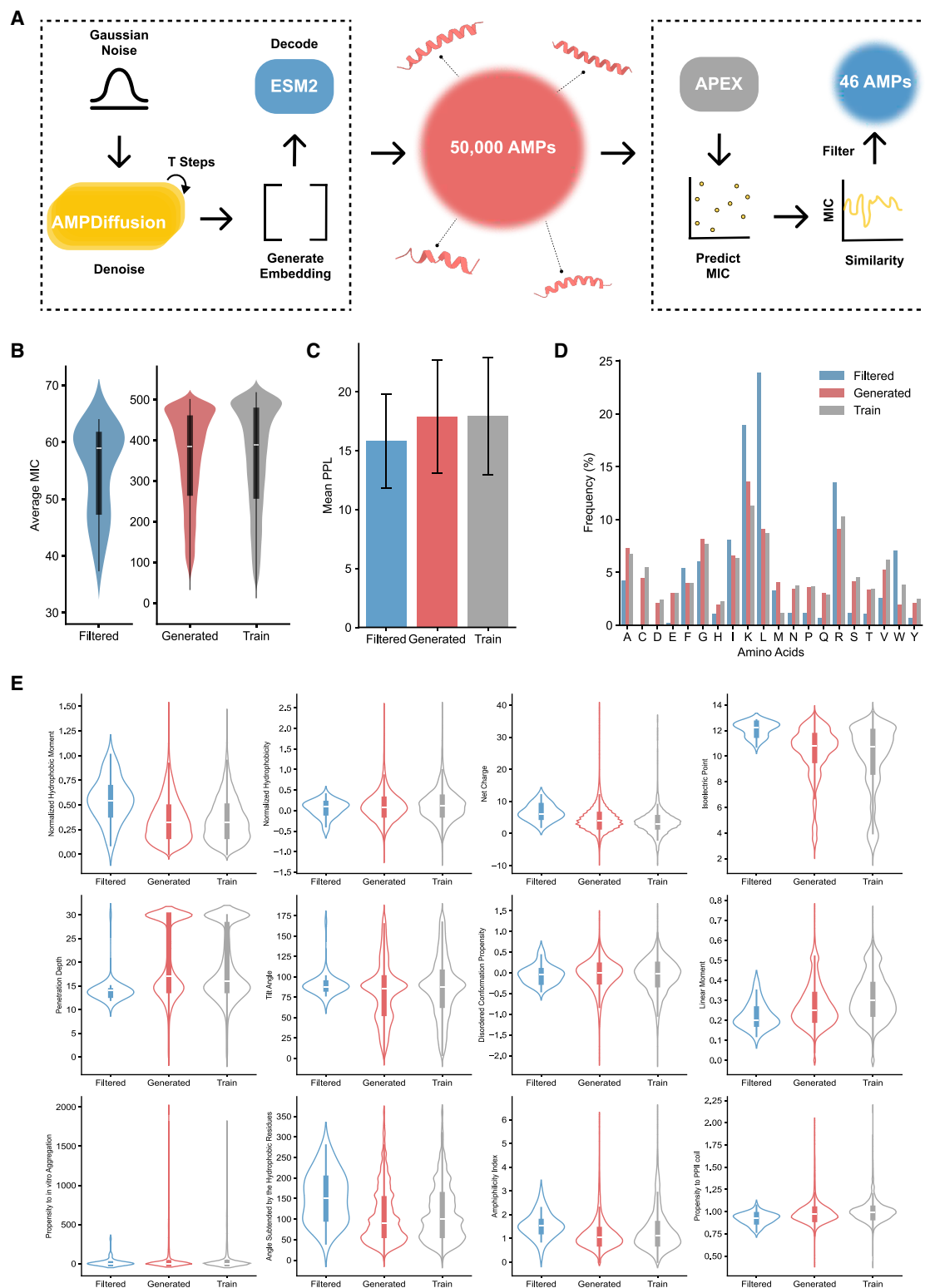
From an initial set of 50,000 AMP candidates (Dataset S2), we selected 46 peptides for experimental validation based on three criteria: high predicted antimicrobial activity, low similarity to existing AMPs, and broad sequence diversity (Figure 1A, see computational filtering in the Methods section for the peptide selection criteria details). The minimum inhibitory concentration (MIC) distribution of our generated sequences was comparable to the training set, indicating the model's ability to recapitulate the MIC profile of training AMPs (Figures 1B and S1). To assess the naturalness of the generated peptides, we employed ProGen2, a pLM trained on millions of natural protein sequences. Naturalness was quantified using perplexity scores, where lower values indicate greater naturalness. Our generated sequences exhibited perplexity values (17.90; filtered 15.83) similar to those of the training set (17.93, two sided Mann-Whitney U test *p*-value 0.0357) (Figure 1C), suggesting that AMP-Diffusion produces peptides with a degree of naturalness comparable to naturally occurring AMPs.

Compositional and physicochemical features of the predicted peptides

The generated AMP library closely resembled the training set in amino acid composition, indicating that our model successfully captured the compositional characteristics of known AMPs (Figure 1D). Interestingly, the filtered subset of generated sequences exhibited a distinct pattern, with notably higher frequencies of lysine (K), leucine (L), and arginine (R). These residues are known to be important for initial electrostatic interactions (K and R) with negatively charged bacterial membranes and for amphiphilicity (K, L, and R), which facilitates effective lipid membrane interactions.^{11,33,34} Thus, enrichment in cationic and aliphatic residues directly influences the physicochemical features of the AMP-Diffusion-predicted peptides (Figures 1E and S2), making the filtered set slightly more amphiphilic and highly charged, with enhanced structure tendencies and increased antimicrobial activity.

In vitro antimicrobial activity

To validate our model's predictive accuracy, we synthesized and experimentally tested 46 peptide sequences. Selection for



(legend on next page)

synthesis was guided by predicted antimicrobial activity and sequence diversity, ensuring broad representation of the generated peptide space (see APEX model and computational filtering in the [Methods](#) section).

Thirty-five (76%) out of 46 synthesized peptides were active against at least one of the 11 clinically relevant bacterial pathogens tested, including members of the ESKAPEE pathogen list¹ ([Figure 2A](#)). Most peptides exhibited broad-spectrum activity, with *A. baumannii* ATCC 19606 and vancomycin-resistant *E. faecium* ATCC 700221 emerging as the most susceptible strains.

Secondary structure

AMPs often adopt an α -helical conformation upon interaction with bacterial membranes.¹¹ This structural transition is influenced by amphipathicity, net charge, hydrophobicity, and peptide length. To assess secondary structure under various biochemical environments, we tested each peptide in water (extracellular environment), a helix-inducing medium (trifluoroethanol in water, 3:2, v/v), a β -inducing medium (methanol in water, 1:1, v/v), and a micelle solution (10 mmol L⁻¹ sodium dodecyl sulfate [SDS]) simulating lipid bilayer interactions.

Consistent with their high lysine (K), leucine (L), and arginine (R) content and in line with known AMPs, the peptides underwent a helix-coil transition. They adopted α -helical conformations in helix-inducing medium and lipid bilayer environments, while remaining unstructured in water and methanol/water mixtures ([Figures 2B–2E](#) and [S3](#)).

Mechanism of action

The bacterial membrane is a common target for most known AMPs due to their non-specific interactions with the lipid bilayer.¹¹ The antimicrobial activity of AMPs is influenced by their amino acid composition, distribution, and physicochemical characteristics such as amphiphilicity and hydrophobicity.

To interrogate the bactericidal mechanism, we first tested whether they permeabilized the outer membrane of *A. baumannii* using 1-(N-phenylamino)naphthalene (NPN) assays. NPN fluoresces more intensely in lipidic than in aqueous environments, allowing the assessment of membrane integrity. All tested peptides were at least as active as the antibiotics polymyxin B and levofloxacin tested in these experiments. The peptides AMP-diff2-1, -13, -32, -33, and -38 showed particularly high levels of outer membrane permeabilization ([Figures 3A](#) and [S4A](#)). These results indicate that the peptides are more effective at permeabilizing bacterial membranes than many previously reported AMPs and encrypted peptides (EPs) derived from human proteins, extinct organisms, or bacterial metagenomes.^{16,18,26,36}

Outer membrane permeabilization is the most common mechanism of action described for AMPs.

Next, we evaluated whether the peptides depolarized the cytoplasmic membrane of *A. baumannii*. Using the potentiometric fluorophore 3,3'-dipropylthiadicarbocyanine iodide (DiSC₃₋₅), which fluoresces upon membrane depolarization, we found that the peptides caused stronger depolarization than polymyxin B and levofloxacin. Their activity was comparable to EPs known for their potent depolarizer properties.¹⁸ AMP-diff2-13, 30, 32, 33, and 42 were the most potent depolarizers ([Figures 3B](#) and [S4B](#)).

Cytotoxicity assays

All 46 synthesized peptides were tested for cytotoxic activity against human embryonic kidney (HEK293T) cells, a well-characterized line commonly used for toxicity.^{33,37,38} Among these, 34 displayed no significant cytotoxicity at the tested concentrations (8–128 μ mol L⁻¹). Cytotoxicity was detected for 12 peptides, and only 6 showed cytotoxicity at concentrations below 64 μ mol L⁻¹ (the maximum concentration tested in bacterial inhibition assays). These findings reinforce the overall excellent safety profiles of this peptide class. The CC₅₀ (peptide dose leading to 50% cytotoxicity) was estimated by non-linear regression ([Figure 3C](#)).

Anti-infective efficacy in an animal model

To assess whether the active peptides had anti-infective efficacy *in vivo*, we used preclinical mouse model of skin abscess infection.^{39–43} Two of the peptides, AMP-diff2-16 and 43, with high *in vitro* activity against *A. baumannii* ATCC 19606 (MIC = 1 μ mol L⁻¹), were tested at a single dose at 10-times MIC (10 μ mol L⁻¹) after infection was established. Each treatment involved a 20 μ L topical injection, corresponding to doses of approximately 24 μ g kg⁻¹ for AMP-diff2-16 and 26 μ g kg⁻¹ for AMP-diff2-43. For comparison, polymyxin B and levofloxacin were also administered at the same concentration (10 μ mol L⁻¹), yielding doses of 13.9 μ g kg⁻¹ and 3.6 μ g kg⁻¹, respectively.

In this model, mice were infected subcutaneously with *A. baumannii* cells ([Figure 4A](#)). A single dose of each peptide was administered directly to the infected area. After two days, both peptides markedly reduced the bacterial load by 1.5 orders of magnitude ([Figure 4B](#)). After four days, bacterial counts had decreased by 2–2.5 orders of magnitude, comparable to reductions achieved by widely used antibiotics polymyxin B and levofloxacin,^{4,8} which were used as antimicrobial controls ([Figure 4B](#)). No weight changes, skin damage, or other adverse effects were observed in treated mice ([Figure 4C](#)).

These substantial *in vivo* results highlight the potential of AMP-Diffusion-generated peptides as effective anti-infective agents,

Figure 1. AMP-Diffusion model and characteristics of the generated peptides

(A) Generation and filtration process. 50,000 sequences were generated from random Gaussian noise using the AMP-Diffusion model.²⁷ Generated Embedding Matrix M is decoded into sequence via ESM2-650M Language Model Head. All sequences underwent minimum inhibitory concentration (MIC) prediction for 11 species using the APEX model. A filtered set of 46 peptides was constructed based on average MIC values (μ mol L⁻¹) and sequence identity criteria. (B) Distribution of average MIC values (μ mol L⁻¹) across filtered, generated, and training sets, presented as a violin plot. (C) Comparison of perplexity (PPL) scores among the three groups (filtered, generated, and training sets), visualized as a bar plot with error bars representing ± 1 standard deviation. (D) Amino acid distribution profiles for the filtered, generated, and training sets. (E) Distribution of key physicochemical features across the three groups.

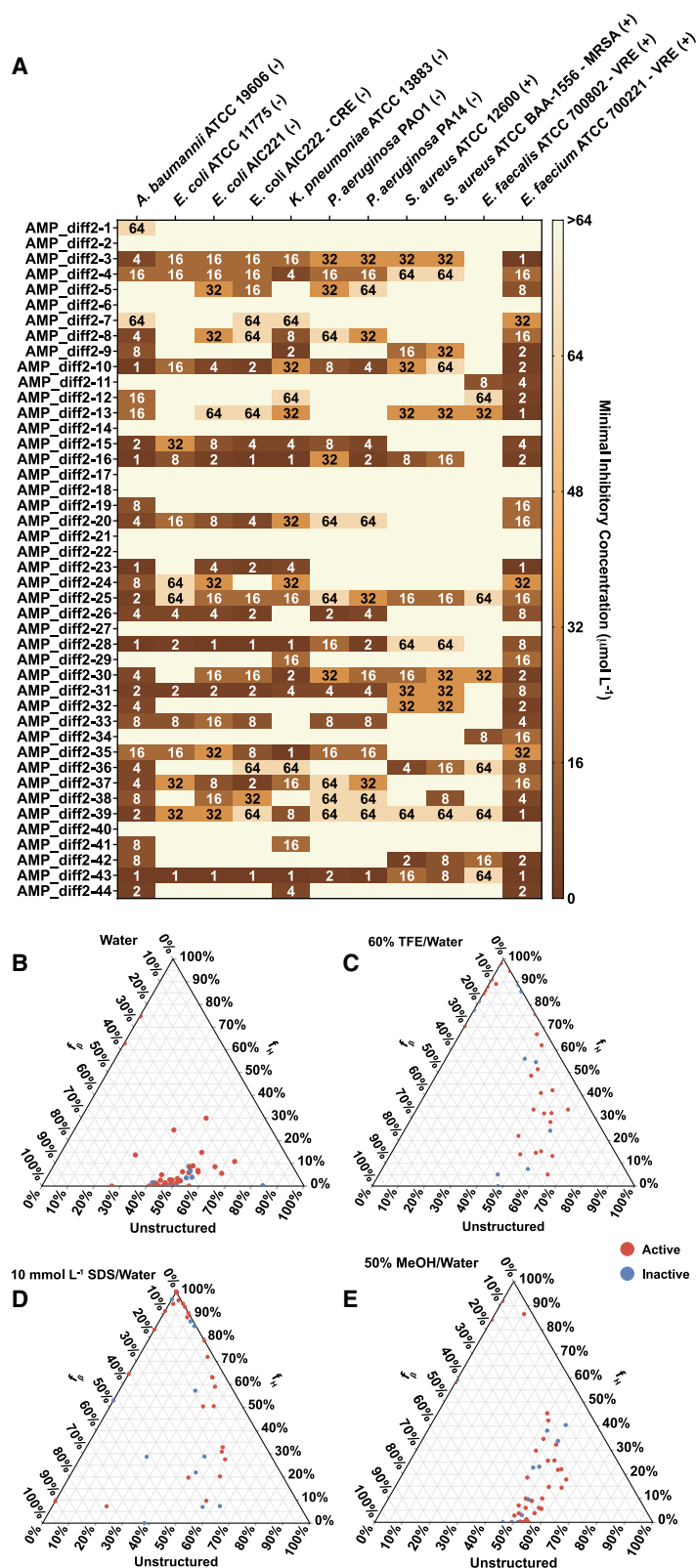


Figure 2. Antimicrobial activity and secondary structure of the peptides generated with AMP-Diffusion

(A) Heatmap of the antimicrobial activities ($\mu\text{mol L}^{-1}$) of the synthesized peptides against 11 clinically relevant pathogens, including four strains resistant to conventional antibiotics. Briefly, 10^6 bacterial cells and serially diluted peptides ($1\text{--}128 \mu\text{mol L}^{-1}$) were incubated at 37°C . One day post-treatment, the optical density at 600 nm was measured in a microplate reader to evaluate bacterial growth in the presence of the peptides. MIC values in the heatmap are the mode of the three replicates in each condition.

(B) Ternary plots showing the percentage of secondary structure for each peptide (at $50 \mu\text{mol L}^{-1}$) in four different solvents: water, 60% trifluoroethanol (TFE) in water, sodium dodecyl sulfate (SDS, 10 mmol L^{-1}) in water, and 50% methanol (MeOH) in water (see also Figure S3). Secondary structure fractions were calculated using the BeStSel server.³⁵ Red dots indicate active peptides, while blue dots represent inactive peptides (see also Figure S4).

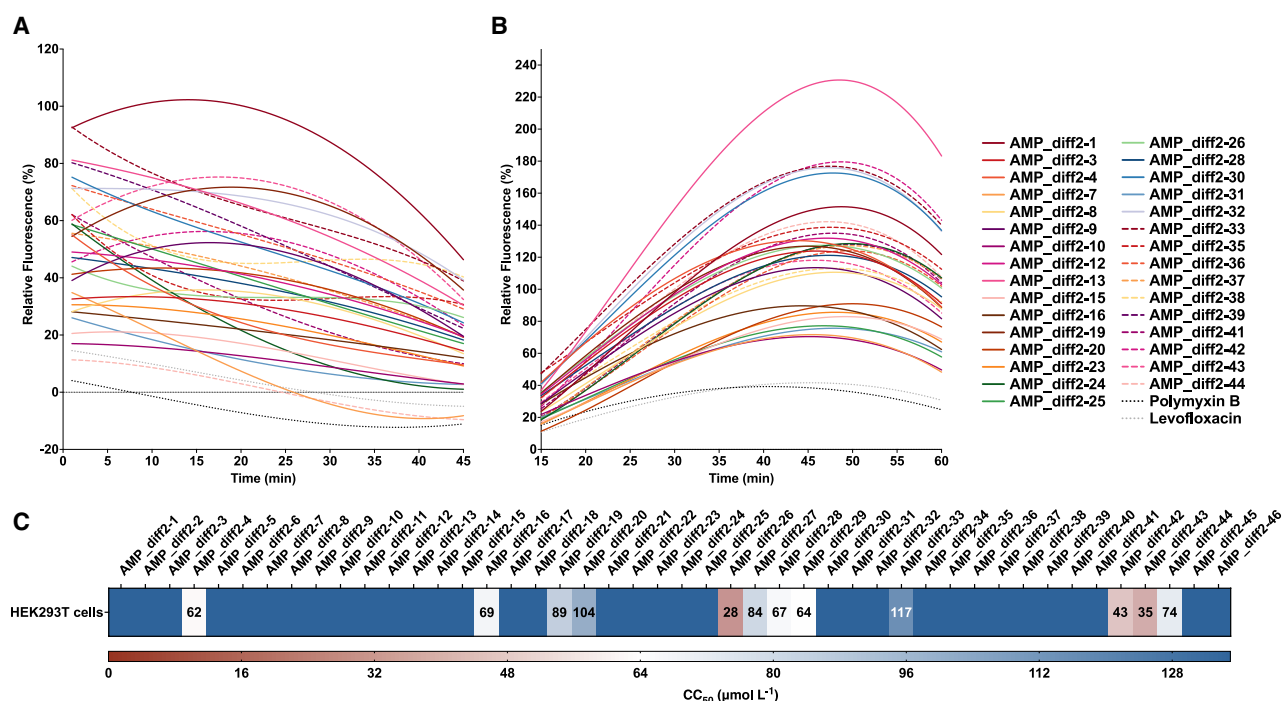


Figure 3. Mechanism of action and cytotoxic activity of the peptides

(A and B) To assess whether the peptides act on bacterial membranes, all active peptides against *A. baumannii* ATCC 19606 were subjected to outer membrane permeabilization and cytoplasmic membrane depolarization assays (see also Figure S4). The fluorescent probe 1-(N-phenylamino)naphthalene (NPN) was used to assess membrane permeabilization (A) induced by the six lead permeabilizer peptides. The fluorescent probe 3,3'-dipropylthiadicarbocyanine iodide (DiSC₃₋₅) was used to evaluate membrane depolarization (B) caused by the six lead depolarizer peptides. The values displayed represent the relative fluorescence of both probes, with non-linear fitting compared with the baseline of the untreated control (buffer + bacteria + fluorescent dye) and benchmarked against the antibiotics polymyxin B and levofloxacin (see also Figure S3 for the results of all peptides with antimicrobial activity against *A. baumannii* ATCC 19606).

(C) Cytotoxic concentrations leading to 50% cell lysis (CC₅₀) were determined by interpolating the dose-response data using a non-linear regression curve. All experiments were performed in three independent replicates.

on par with standard-of-care antibiotics under physiologically relevant conditions.

Limitations of the study

The current version of AMP-Diffusion is based on the original denoising diffusion probabilistic models (DDPM) framework,¹ which has not been optimized for generation quality and computational efficiency compared with more recent methods like elucidated diffusion models (EDM⁴⁴ and EDM2⁴⁵). In addition, AMP-Diffusion operates as an unconditional diffusion model, lacking conditional guidance mechanisms. Future work may explore training dedicated classifiers for noised sequences to guide generation based on key therapeutic properties (e.g., inhibitory activity and toxicity), or, alternatively, using classifier-free guidance with pathogen-specific and therapeutic property embeddings to produce peptides tailored for particular bacterial strains with high safety. We believe that discrete diffusion architectures, such as masked discrete diffusion (MDLM) and discrete denoising posterior prediction (DDPP), will be apt architectural candidates for a next-generation AMP-Diffusion model.^{46–48} Incorporating these innovations could yield a next-generation framework that further refines peptide design for enhanced antimicrobial efficacy and safety. Publicly available

AMP databases are characterized by significant compositional heterogeneity, encompassing peptides that act on a wide array of target groups and molecular objects. For example, in the DBAASP database, there is a clear imbalance in research focus, with a predominant number of peptides targeting gram-positive and gram-negative bacteria compared with other organisms like fungi, viruses, parasites, or cancer cells. This diversity extends beyond cellular targets to include non-biological structures, such as the lipid bilayer, which is a common target for membrane-disrupting peptides. While this breadth provides a comprehensive view of the AMP landscape, the inherent imbalance can pose a challenge for developing unbiased computational models. For future studies where this heterogeneity is a significant concern, this could be effectively addressed through strategic subsampling to create more balanced and focused datasets tailored to specific organisms or mechanisms of action.

METHODS

AMP-Diffusion

AMP-Diffusion is a latent-space diffusion model designed for generating AMPs following the DDPM framework. It leverages the state-of-the-art ESM-2 pLM to map peptide sequences

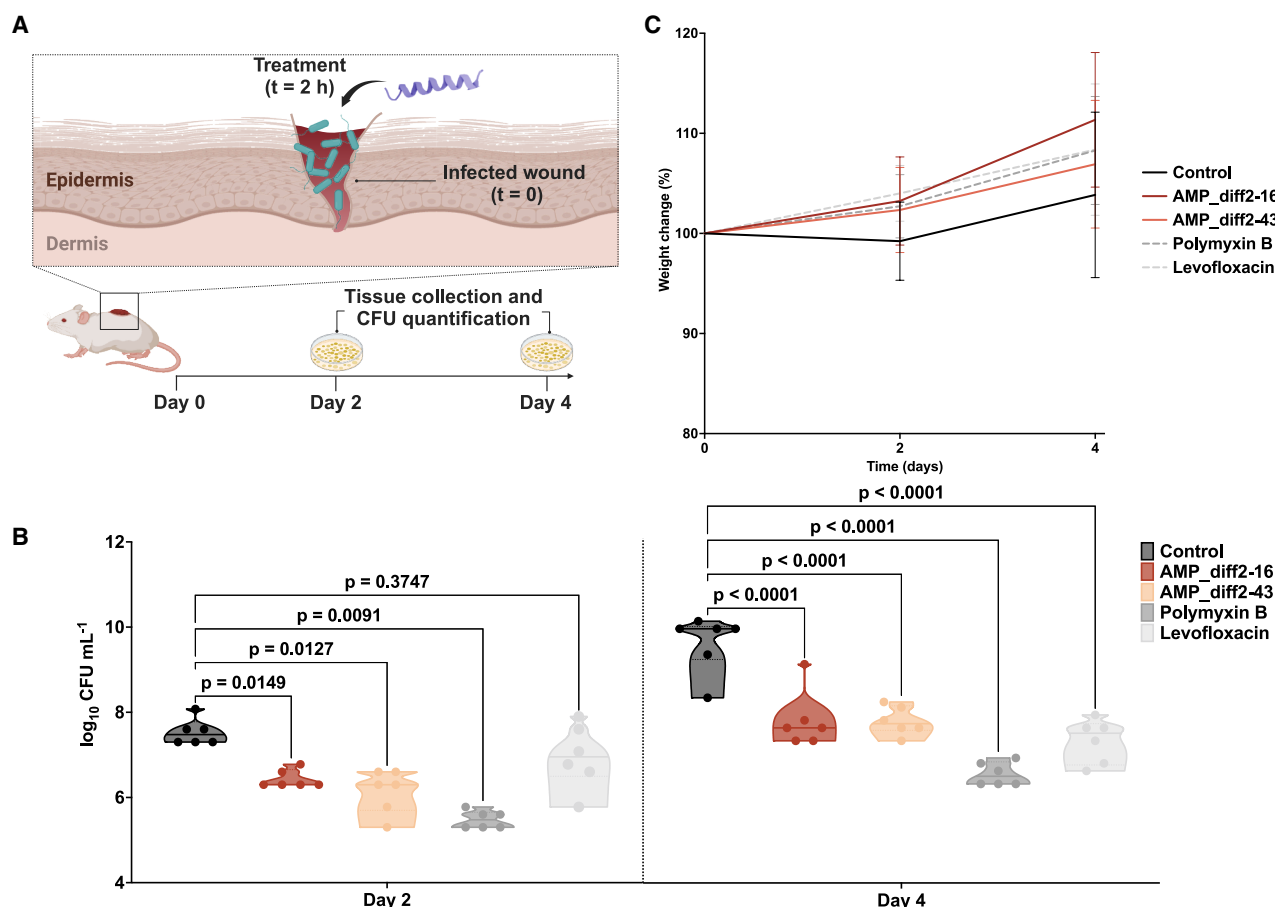


Figure 4. Anti-infective activity of the peptides in a murine infection model

(A) Schematic of the skin abscess mouse model used to assess selected peptides ($n = 6$) for activity against *A. baumannii* ATCC 19606.

(B) The peptides AMP-diff2-16 and 43, administered at $10 \mu\text{mol L}^{-1}$ in a single dose, each treatment was made as a $20 \mu\text{L}$ topical injection, corresponding to doses of approximately $24 \mu\text{g kg}^{-1}$ for AMP-diff2-16 and $26 \mu\text{g kg}^{-1}$ for AMP-diff2-43. For comparison, polymyxin B and levofloxacin were also administered at the same concentration ($10 \mu\text{mol L}^{-1}$), yielding doses of $13.9 \mu\text{g kg}^{-1}$ and $3.6 \mu\text{g kg}^{-1}$, respectively. The peptides significantly inhibited bacterial proliferation for up to four days post-treatment compared with the untreated control group. Both peptides reduced the infection to a level comparable with the antibiotic controls polymyxin B and levofloxacin.

(C) Mouse weight was monitored throughout the duration of the skin abscess model (4 days total) to assess potential toxicity from bacterial load or peptide treatment. Statistical significance in B was determined using one-way ANOVA followed by Dunnett's test; p values are shown in the graph. Figure A was created with BioRender.com.

into a continuous latent space, where Gaussian noise is systematically added during the forward process. The model then trains a denoiser to reconstruct the original latent embeddings from the noised inputs, minimizing the l2 loss between predicted and original latent embeddings. For generation, the reverse process starts with Gaussian noise and iteratively denoises it to produce samples resembling the original data distribution. The denoising architecture employs pre-trained ESM-2-8M attention blocks integrated with positional time embeddings and a multilayer perceptron (MLP) for final processing. For the purposes of this project, we retrained the model on a clean and expanded dataset (size = 19,670 peptides, Dataset S1) from DRAMP, DBAASP, and APD3 using exponential moving average. That is, along with the diffusion model training, we maintain a separate, "slow-moving average" copy of the model's weights throughout

the training process. After each training step, where the primary model's weights are updated by the optimizer, the Exponential Moving Average (EMA) weights are updated using a small fraction of the new weights. The update rule is as follows: $\Theta_{\text{ema},t} = \text{decay} * \Theta_{\text{ema},t-1} + (1 - \text{decay})\Theta_{\text{current},t}$, where $\Theta_{\text{ema},t}$ is the model's EMA weights at training step t , $\Theta_{\text{current},t}$ is the model's weights after the optimizer update at step t and decay is a hyperparameter set to a value very close to 1. In our case, decay is set to 0.995. This ensures that the EMA weights evolve very slowly and smoothly over time. For generation, we use this final set of EMA weights rather than the weights from the very last training step. All other parameters and methodologies remained consistent with the original model architecture. For a comprehensive understanding of the technical details, it is encouraged to refer to the original research.

APEX model

We utilized APEX,⁴⁹ a bacterial strain-specific antimicrobial activity predictor for peptide sequences to rank and select the generated peptides (<https://gitlab.com/machine-biology-group-public/apex-pathogen>). As APEX predicts antimicrobial activities against eleven pathogen strains, we used the mean prediction MIC to sort the peptides obtained by AMP-Diffusion. Top-scored peptides that were not filtered out by our diversity and similarity criteria (see computational filtering section) were selected for synthesis and validation.

Computational filtering

To ensure that the selected sequences for validation were (a) showing high antimicrobial activities, (b) sufficiently distinct from known AMPs, and (c) covered a diverse sequence space, we applied three filtering criteria to the generated peptide sequences: (1) the generated peptides must have mean MIC $\leq 64 \mu\text{mol L}^{-1}$. (2) Similarity to known AMPs, where we calculated sequence similarity for each generated peptide using local sequence alignment¹⁸ against a curated set of AMPs from DRAMP 3.0,³⁰ DBAASP,³² and APD3,³¹ along with peptides from our inventory. Peptides with a sequence similarity > 0.60 to any known AMP were considered too similar and were excluded from further analysis. (3) Diversity of remaining peptides, in which, for the remaining peptides, we performed pairwise sequence similarity comparisons. When two peptides had a sequence similarity > 0.40 , we retained only the one with the lower predicted mean MIC, as estimated by APEX.¹⁸

ProGen2 and perplexity

ProGen2 is a suite of pLMs developed for protein sequence generation and understanding. The model is based on the autoregressive language model. ProGen2 comes in various sizes, ranging from 151 million to 6.4 billion parameters, allowing for different computational requirements and applications. The models use a standard transformer decoder with left-to-right causal masking and incorporate rotary positional encodings. ProGen2 was trained on a mixture of protein sequences from diverse sources including UniRef90, UniProtKB, and BFD30. This diverse training data allows ProGen2 to learn patterns and relationships across a wide range of protein families and structures.

Perplexity measures how well a probabilistic model predicts a sample and is commonly used in language models, including pLMs, to evaluate sequence quality. The perplexity is defined as the exponential of the average negative log-likelihood per token.

$$\text{PPL} = \exp\left(-\frac{1}{N} \sum_{i=1}^N \log P(x_i)\right)$$

where N is the number of tokens, and $P(x_i)$ is the predicted probability of the i th token. Lower perplexity indicates better model confidence. For pLMs like ProGen2, perplexity is associated with the model's ability to capture the distribution of observed evolutionary sequences. In our study, we utilized the ProGen2-medium for scoring. Lower perplexity suggests a higher likelihood of our model producing valid and biologically natural protein sequences.

Physicochemical features

Generated peptides from AMP-Diffusion were compared with natural AMPs (the training data). The physicochemical properties, including normalized hydrophobic moment, normalized hydrophobicity, net charge, isoelectric point, penetration depth, tilt angle, disordered conformation propensity, linear moment, propensity to *in vitro* aggregation, angle subtended by the hydrophobic residues, amphiphilicity index, and propensity to polyproline type II (PPII) coil, were calculated using the DBAASP server.³²

Peptide synthesis

Peptides were obtained from AAPPTec and synthesized by solid-phase peptide synthesis using Fmoc strategy.

Bacterial strain and culture

The strains used in the MIC determination and mechanism of action experiments were the following: *Acinetobacter baumannii* ATCC 19606, *Escherichia coli* ATCC 11775, *Escherichia coli* AIC221 (*Escherichia coli* MG1655 phnE_2::FRT [control strain for AIC222]) and *Escherichia coli* AIC222 (*Escherichia coli* MG1655 pmrA53 phnE_2::FRT [polymyxin resistant; colistin-resistant strain]), *Klebsiella pneumoniae* ATCC 13883, *Pseudomonas aeruginosa* PAO1, *Pseudomonas aeruginosa* PA14, *Pseudomonas aeruginosa* ATCC 35032, *Staphylococcus aureus* ATCC 12600, *Staphylococcus aureus* ATCC 6538, *Staphylococcus aureus* ATCC BAA-1556 (methicillin-resistant strain), *Enterococcus faecalis* ATCC 700802 (vancomycin-resistant strain), and *Enterococcus faecium* ATCC 700221 (vancomycin-resistant strain). All strains were grown and plated on Luria-Bertani (LB), *Pseudomonas* Isolation (*Pseudomonas aeruginosa* strains), and MacConkey (*A. baumannii* strain) agar plates and incubated overnight at 37°C from frozen stocks. After incubation, one isolated colony was transferred to 5 mL of LB medium, and cultures were incubated overnight (16 h) at 37°C. The following day, inocula were prepared by diluting the overnight cultures 1:100 in 5 mL of the respective media and incubating them at 37°C until bacteria reached logarithmic phase (optical density 600 (OD_{600}) = 0.3–0.5).

MIC determination

Broth microdilution assays⁵⁰ were conducted to establish the MIC for each peptide. Peptides were first added to untreated polystyrene 96-well microtiter plates and then serially diluted two-fold in sterile water, with final concentrations ranging from 1 to 64 $\mu\text{mol L}^{-1}$. A bacterial inoculum, prepared at a density of 10^6 colony-forming units (CFU) mL^{-1} in LB medium, was mixed with the peptide solutions at a 1:1 ratio. Following a 24-h incubation at 37°C, the MIC was defined as the lowest peptide concentration that completely inhibited bacterial growth. Each assay was conducted in triplicate across three independent experiments.

Outer membrane permeabilization assays

The NPN uptake assay was employed to assess the ability of peptides to permeabilize the bacterial outer membrane. Cultures of *A. baumannii* ATCC 19606 were grown to an OD_{600} of

0.4, centrifuged at 10,000 rpm for 10 min at 4°C, washed, and resuspended in 5 mmol L⁻¹ HEPES buffer (pH 7.4) supplemented with 5 mmol L⁻¹ glucose. The bacterial suspension (100 µL) was added to a white 96-well plate, followed by the addition of 4 µL NPN at a final concentration of 0.5 mmol L⁻¹. Peptides diluted in water were subsequently added to each well, and fluorescence measurements were recorded over 45 min at an excitation wavelength (λ_{ex}) of 350 nm and an emission wavelength (λ_{em}) of 420 nm. Relative fluorescence was calculated using an untreated control (buffer + bacteria + fluorescent dye) as the baseline, with the percentage difference between the baseline and sample values determined using the following equation:

$$\text{Percentage difference} = \frac{100 * (\text{fluorescence}_{\text{sample}} - \text{fluorescence}_{\text{untreated control}})}{\text{fluorescence}_{\text{untreated control}}}$$

Cytoplasmic membrane depolarization assays

The cytoplasmic membrane depolarization assay utilized the membrane potential-sensitive dye DiSC₃-5 to evaluate peptide activity. Mid-logarithmic phase cultures of *A. baumannii* ATCC 19606 were washed and resuspended in HEPES buffer (pH 7.2) containing 20 mmol L⁻¹ glucose and 0.1 mol L⁻¹ KCl at a concentration of 0.05 OD₆₀₀ mL⁻¹. DiSC₃-5 was added to the bacterial suspension at a final concentration of 20 µmol L⁻¹ and incubated for 15 min to allow stabilization of fluorescence, indicating dye incorporation into the bacterial membrane. Peptides were then mixed with the bacterial suspension at a 1:1 ratio to achieve final concentrations equivalent to their MIC values. Membrane depolarization was monitored by measuring fluorescence changes (λ_{ex} = 622 nm, λ_{em} = 670 nm) over 60 min. Relative fluorescence was calculated using an untreated control (buffer + bacteria + dye) as the baseline, and the percentage difference between the baseline and the sample was determined using the following equation:

$$\text{Percentage difference} = \frac{100 * (\text{fluorescence}_{\text{sample}} - \text{fluorescence}_{\text{untreated control}})}{\text{fluorescence}_{\text{untreated control}}}$$

Circular dichroism

Circular dichroism (CD) experiments were carried out using a J1500 CD spectropolarimeter (Jasco) at the Biological Chemistry Resource Center (BCRC) of the University of Pennsylvania. Measurements were conducted at 25°C, with spectra representing the average of three accumulations. A quartz cuvette with a 1.0 mm optical path length was used, and data were collected from 260 to 190 nm at a scanning rate of 50 nm min⁻¹ with a bandwidth of 0.5 nm. Peptides were tested at a concentration of 50 µmol

L⁻¹ in four different solvent conditions: water, a 3:2 trifluoroethanol (TFE)/water mixture, 10 mmol L⁻¹ SDS in water, and a 1:1 methanol (MeOH)/water mixture, with corresponding baselines recorded prior to measurement. A Fourier transform filter was applied to reduce background noise. Secondary structure fractions were determined using the BeStSel³⁵ single spectra analysis tool. Ternary plots were generated using <https://www.ternaryplot.com/> and subsequently modified for visualization.

Eukaryotic cell culture conditions and cytotoxicity assays

HEK293T cells were obtained from the American Type Culture Collection (CRL-3216). The cells were cultured in high-glucose Dul-

becco's modified Eagle's medium supplemented with 1% penicillin and streptomycin (antibiotics) and 10% fetal bovine serum and grown at 37°C in a humidified atmosphere containing 5% CO₂.

One day before the experiment, an aliquot of 100 µL of the cells at 50,000 cells per mL was seeded into each well of the cell-treated 96-well plates used in the experiment (that is, 5,000 cells per well). The attached HEK293T cells were then exposed to increasing concentrations of the peptides (8–128 µmol L⁻¹) for 24 h. After the incubation period, we performed the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide tetrazolium reduction assay (MTT assay). The MTT reagent was dissolved at 0.5 mg mL⁻¹ in medium without phenol red and was used to replace cell culture supernatants containing the peptides (100 µL per well), and the samples were incubated for 4 h at 37°C in a humidified atmosphere containing 5% CO₂ yielding the insoluble formazan salt. The resulting salts were then resuspended in hydrochloric acid (0.04 mol L⁻¹) in anhy-

drous isopropanol and quantified by spectrophotometric measurements of absorbance at 570 nm. All assays were done as three biological replicates.

Skin abscess infection mouse model

Six-week-old female CD-1 mice were anesthetized, and their backs were shaved. The area was then wiped with ethanol to eliminate any endogenous bacteria prior to creating a surface abrasion using a scalpel. An aliquot of *A. baumannii* ATCC

19606 (10^6 CFU mL⁻¹; 20 μ L), grown to an OD of 0.5 at 600 nm in LB medium and washed twice with sterile PBS (pH 7.4) by centrifugation ($5,000 \times g$ for 3 min), was applied to the scratched region. Peptides, polymyxin B or levofloxacin, each diluted to the $10 \times$ MIC, was administered to the wound site 2 h post-infection. Two and four days after the infection, the animals were euthanized, and the infected skin was excised. The tissues were homogenized using a bead beater at 25 Hz for 20 mins. The samples were then serially diluted 10-fold and plated on MacConkey agar plates for CFU quantification. Six mice per group were included in the study. All procedures were approved by the University Laboratory Animal Resources (ULAR) at the University of Pennsylvania (protocol 806763).

Statistical analyses

The data shown represent the median and upper and lower quartiles from a minimum of three independent experiments. Statistical significance in Figure 4 was calculated using a one-way ANOVA for single comparisons. All statistical analyses were performed using GraphPad Prism v.10.0.3. Significant differences are expressed as *p* values.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled upon reasonable request by the lead contact, Cesar de la Fuente-Nunez (cfuente@upenn.edu). Additional requests for AMP-Diffusion data or model support will be fulfilled by corresponding author, Pranam Chatterjee (pranam@seas.upenn.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All training data, testing data, and code used to develop the AMP-Diffusion are freely available on GitHub (<https://github.com/programmablebio/amp-diffusion>).
- Datasets are available in Mendeley Data (<https://data.mendeley.com/datasets/ybh43wf43g/1>).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon reasonable request.

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AUTHOR CONTRIBUTIONS

L.T.C. and P.C. conceived, designed, and developed AMP-Diffusion architecture, training, and analysis. P.C. directed and supervised the development of

AMP-Diffusion. M.D.T.T. and C.d.I.F.-N. conceived and designed the experimental validation of peptides generated with AMP-Diffusion. M.D.T.T. performed all experiments. F.W. performed the computational filtering, physicochemical feature calculation, and peptide visualization. C.F.N. supervised the experiments. M.D.T.T., L.T.C., and F.W. performed data analysis. C.d.I.F.-N. and P.C. funded the project. All authors wrote the original draft and revised the final version.

DECLARATION OF INTERESTS

C.d.I.F.-N. is a co-founder of, and scientific advisor, to Peptaris, Inc., provides consulting services to Invaio Sciences, and is a member of the Scientific Advisory Boards of Nowture S.L., Peptidus, European Biotech Venture Builder, the Peptide Drug Hunting Consortium (PDHC), ePhectic Therapeutics, Inc., and Phare Bio. P.C. is a co-founder and scientific advisor to Gameto Inc. and UbiquiTx Inc., and is a member of the Scientific Advisory Boards of Latus Bio, Inc., DreamFold Inc., Juniper Genomics Inc., and Atom Bioworks Inc. M.D.T.T. is a co-founder and scientific advisor to Peptaris Inc. All the other authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the author(s) used ChatGPT in order to review grammar. After using this tool or service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

SUPPLEMENTAL INFORMATION

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